

Connecting a buzzer directly to a Raspberry Pi 5 GPIO pin causes continuous beeping because the Pi's pins operate at 3.3V and can only safely supply about 16mA of current. Direct connections either pull the pin into a floating state or draw too much current, which risks permanently burning out the Raspberry Pi's processor.

To safely switch a 5V buzzer using a 3.3V Raspberry Pi signal, you must use an **NPN Transistor** acting as a digital switch, paired with a **Current-Limiting Resistor**.

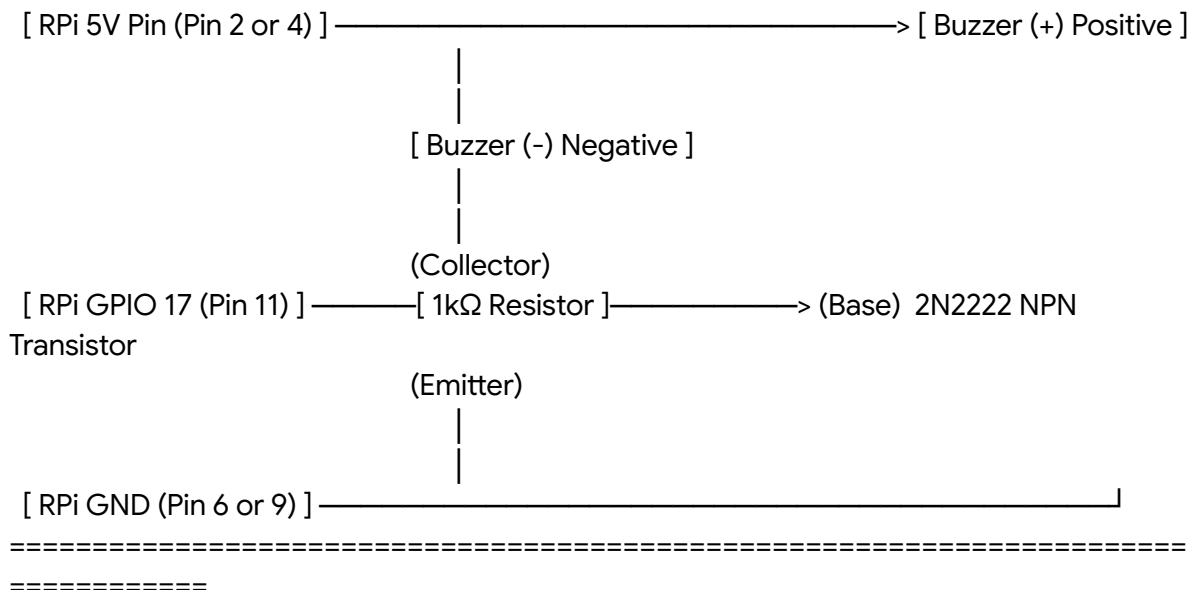
Required Components

- **1x NPN Transistor (2N2222 or BC547):** Acts as the heavy-duty switch to route 5V power to the buzzer without touching the Pi's delicate logic pins.
- **1x $1\text{k}\Omega$ Resistor:** Protects the Raspberry Pi GPIO pin by limiting the current flowing into the transistor base.
- **1x 5V Active Buzzer:** Generates sound automatically when 5V DC is applied.
- *(Optional)* **1x 1N4007 Diode:** Wired across the buzzer to absorb reverse-voltage spikes (highly recommended if the buzzer is magnetic rather than piezoelectric).

Raspberry Pi 5 Buzzer Wiring Schematic

Plaintext

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RASPBERY PI 5 SAFE BUZZER DRIVER CIRCUIT
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```



How it Works:

When the Raspberry Pi sends a HIGH (3.3V) signal from GPIO 17, it passes safely through the Ω resistor and turns the transistor "ON". The transistor then completes the circuit to Ground, allowing the 5V power from the Pi to flow through the buzzer and make it scream.

Python Control Script (gpiozero)

Since you are using a Raspberry Pi 5, the modern standard for GPIO control is the gpiozero library. This script handles both the single door beep and the continuous emergency siren.

```
Python
from gpiozero import Buzzer
from time import sleep

# Initialize the buzzer on GPIO 17
alarm_buzzer = Buzzer(17)

def door_beep():
    """Single 100ms beep for door open/close events."""
    print("[EVENT] Main Door Toggled")
    alarm_buzzer.on()
    sleep(0.1)
    alarm_buzzer.off()

def emergency_alarm():
    """Continuous pulsing alarm for Fire/Earthquake."""
    print("💣 [ALARM] EMERGENCY DETECTED! 💣")
    try:
        while True:
            alarm_buzzer.on()
            sleep(0.5)
            alarm_buzzer.off()
            sleep(0.5)
    except KeyboardInterrupt:
        alarm_buzzer.off()
        print("Alarm silenced.")

# --- Test the functions ---
if __name__ == '__main__':
    # Test door beep
    door_beep()
```

```
sleep(2)
```

```
# Trigger emergency alarm  
emergency_alarm()
```